Probability with Engineering Applications: Week4

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Announcement

- 1st Exam: **10:30-12:00, Oct 22**
- You may not use any books, electronic devices, or notes other than **one two-sided sheet of handwritten A4 paper**.
- Content: Week 1 to Week 4 (this week)
- Location: to be determined, will be announced on BB

Programme

- Bernoulli process and negative binomial distribution
- Continuous random variables
- Continuous distributions

Bernoulli process

Negative binomial distribution

Continuous distributions

Continuous random variables

Definition:
A random variable X is *continuous* if

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx$$

for all values $a \leq b$ and a function $f : \mathbb{R} \to [0, \infty)$.

- f is the *probability density function* of X
- the distribution function of X is $F(x) = P(X \leq x) = \int_{-\infty}^{x} f(x)\,dx$
- $f(x) = \frac{d}{dx} F(x)$

Continuous distribution

Let f be the probability density function and F the distribution function of a continuous random variable X. Consider the following two statements:

1. $P(a \leq X \leq b) = F(b) - F(a)$
2. $P(X = a) = f(a)$

Which of these statements is/are true?

A. None

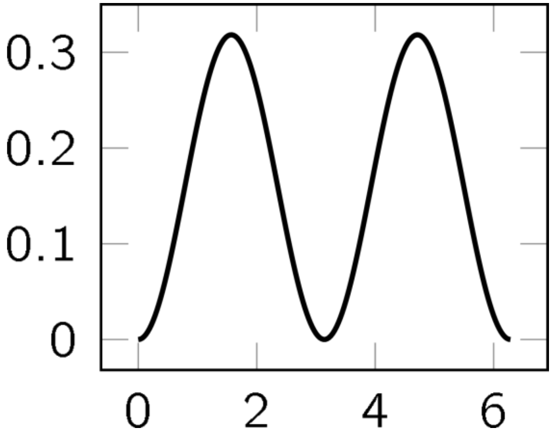
B. Only 1

C. Only 2

D. Both

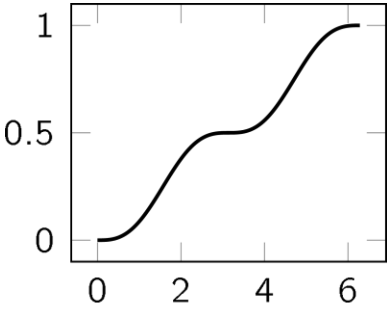
Distribution function

Consider the following probability density function.

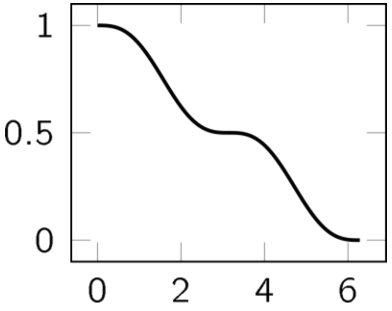


Which is the corresponding distribution function?

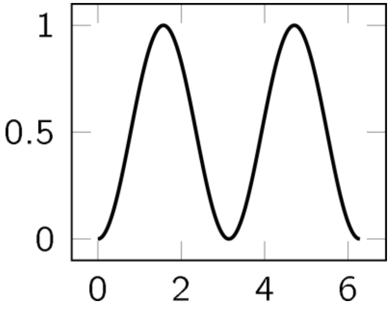
A.



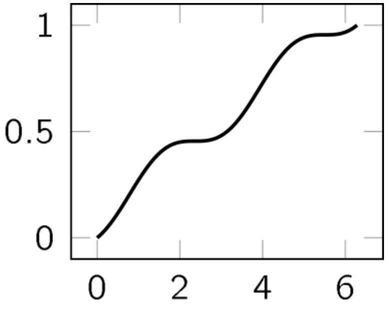
B.



C.



D.



Four standard continuous distributions

Bitcoin: mining times

In the bitcoin network, miners compete to add a block to the blockchain (called mining a block).

We record every day at which times in hours a new block is added, so the outcomes are in $[0, 24]$. Bitcoin is designed such that additions to the blockchain are equally likely to occur in every time interval with a fixed length.

What is the distribution of the mining time?

The uniform distribution

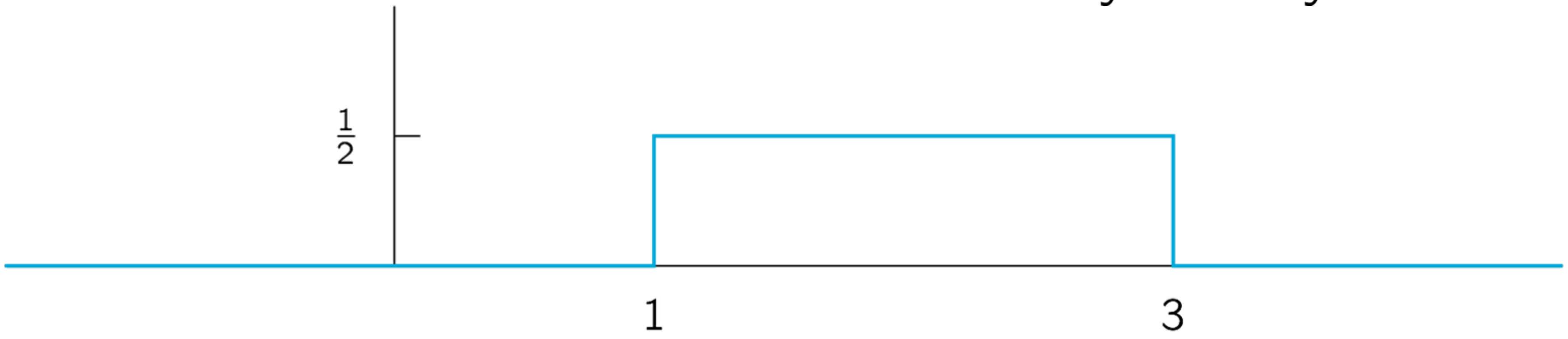
Definition:
A continuous random variable X has a uniform distribution on the interval $[\alpha, \beta]$ if its probability density function f is given by:

$$f(x) = \begin{cases} \dfrac{1}{\beta - \alpha} & \text{for } x \in [\alpha, \beta] \\ 0 & \text{for } x \text{ not in } [\alpha, \beta] \end{cases}$$

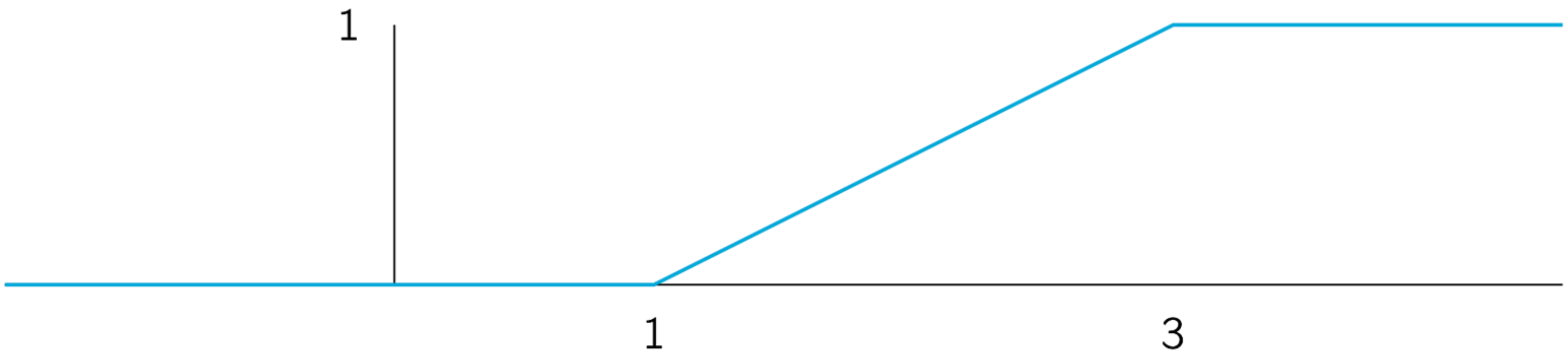
Notation: $X \sim U(\alpha, \beta)$.

Uniform distribution: $U(1, 3)$

Probability density function



Distribution function



Bitcoin: time until mining the next block

Bitcoin is designed such that the mining process is highly unpredictable.

- The average duration until the next block is mined is (roughly) 10 minutes.
- Knowing how much time has elapsed since the previous block was mined gives you no information about the time until the next block will be mined.

What distribution has these properties?

The exponential distribution

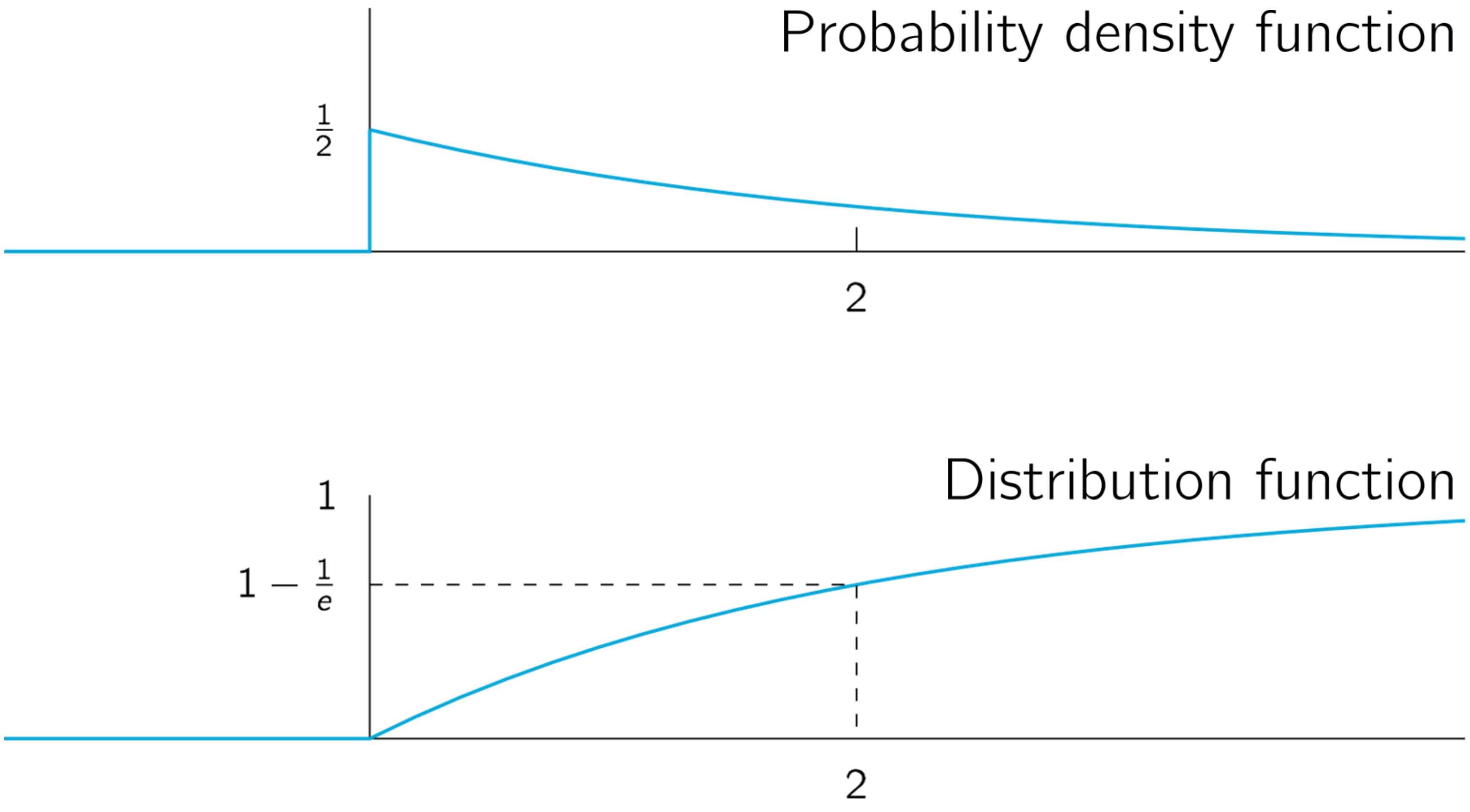
Definition:
A continuous random variable X has an exponential distribution with parameter λ if its probability density function f is given by:

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$$

Notation: $X \sim \text{Exp}(\lambda)$.

The exponential distribution often arises in the context of waiting times, such as the time until a radioactive particle decays, the next earth quake occurs, or a new visitor arrives at a website.

The exponential distribution: $\mathrm{Exp}\left(\frac{1}{2}\right)$



The exponential distribution

The exponential distribution occurs in applications such as reliability theory and queuing theory. Reasons for its use include:

- Its memoryless (Markov) property
- Its relation to the Poisson process

The following random variables will often be modeled as exponential:

- Time between two successive job arrivals to a computing center
- Service time at a server in a queuing network
- Time to failure (lifetime) of a component
- Time required to repair a component that has malfunctioned

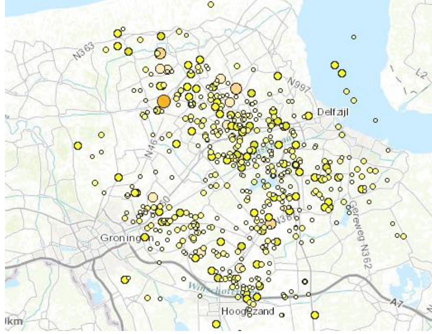
Memoryless Property

- If the component lifetime is memoryless, then the distribution of the remaining life of the component does not depend on how long the component has been operating; the component does not “age.” Its eventual breakdown results from a suddenly appearing failure, not from gradual deterioration.

- If inter-arrival times are exponentially distributed, the “memoryless” property (also known as the Markov property) says that “the wait time for a new arrival is statistically independent of how long we have already waited”!

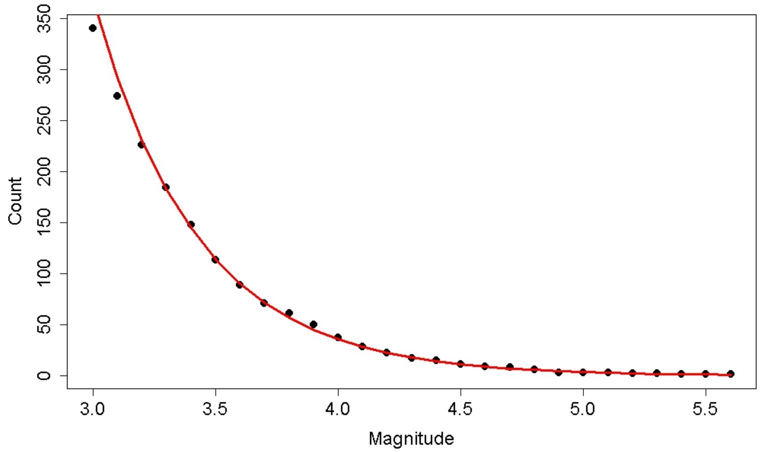
Relation between exponential and geometric distributions

Earthquakes



In the Dutch province Groningen (north east), earthquakes occur due to gas production in this area. Plotting the number of earthquakes versus their magnitude yields:

How to describe this?



M magnitude of an earthquake, then for a certain α it holds that: $P(M > x) = x^{-\alpha}$.

This is called the Gutenberg-Richter law.

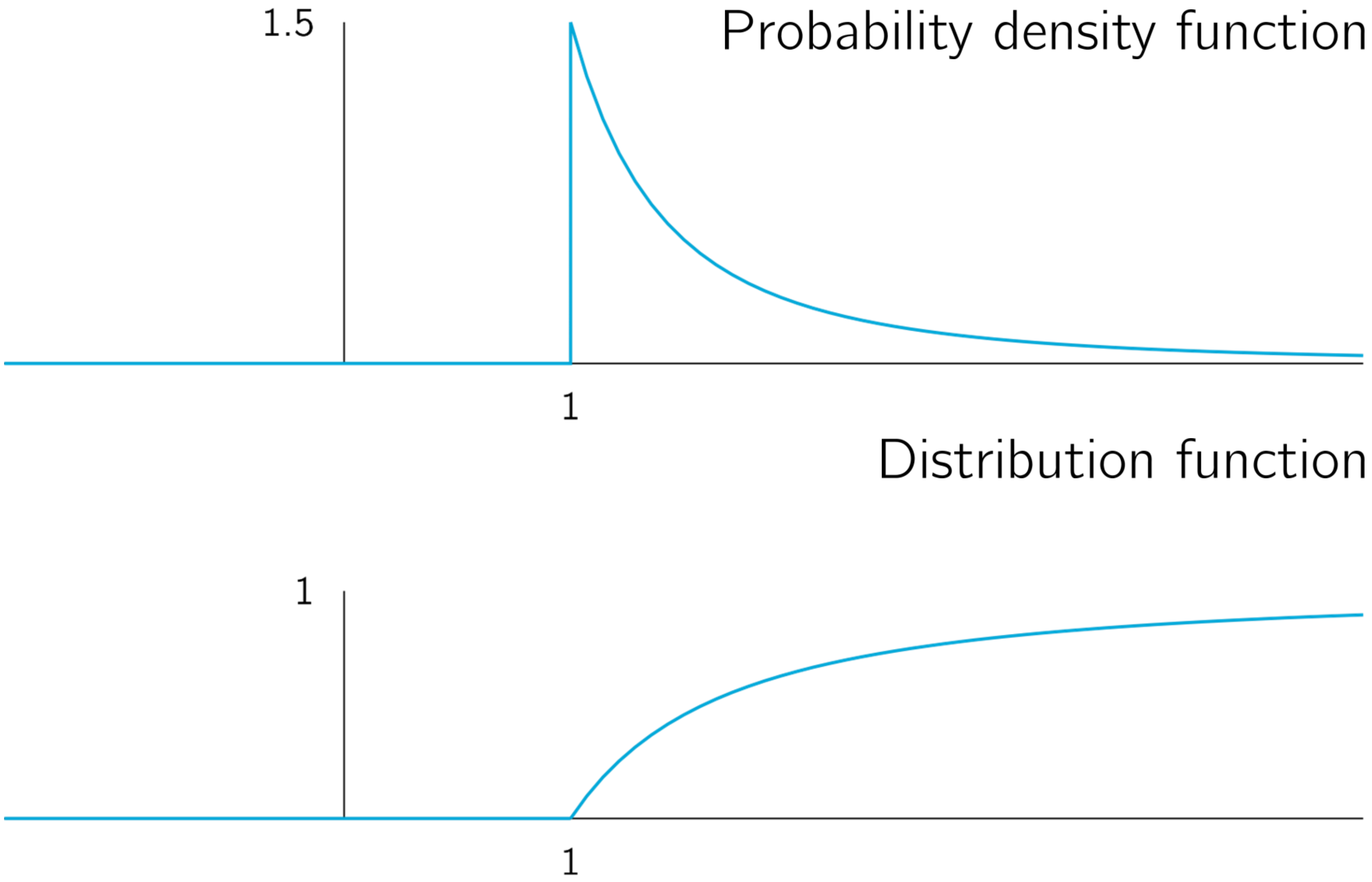
The Pareto distribution

Definition:
A continuous random variable X has a Pareto distribution with parameter $\alpha > 0$ if its probability density function f is given by:

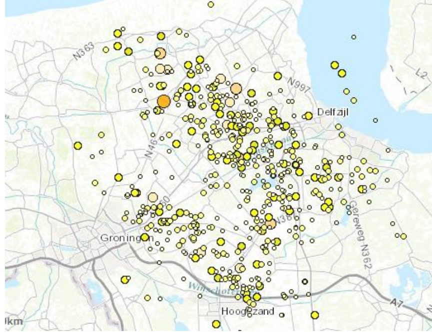
$$f(x) = \begin{cases} \dfrac{\alpha}{x^{\alpha+1}} & \text{for } x \geq 1 \\ 0 & \text{for } x < 1 \end{cases}$$

Notation: $X \sim \mathrm{Par}(\alpha)$.

Pareto distribution: $\mathrm{Par}\left(\frac{3}{2}\right)$



Earthquakes



In the Dutch province Groningen (north east), earthquakes occur due to gas production in this area. If an earthquake occurs, a seismic station measures its magnitude.

The measurement is never perfect, due to imperfect instruments and environmental noise for instance. The total measurement error is the accumulation of many small measurement errors.

What distribution describes the total measurement error?

The normal distribution

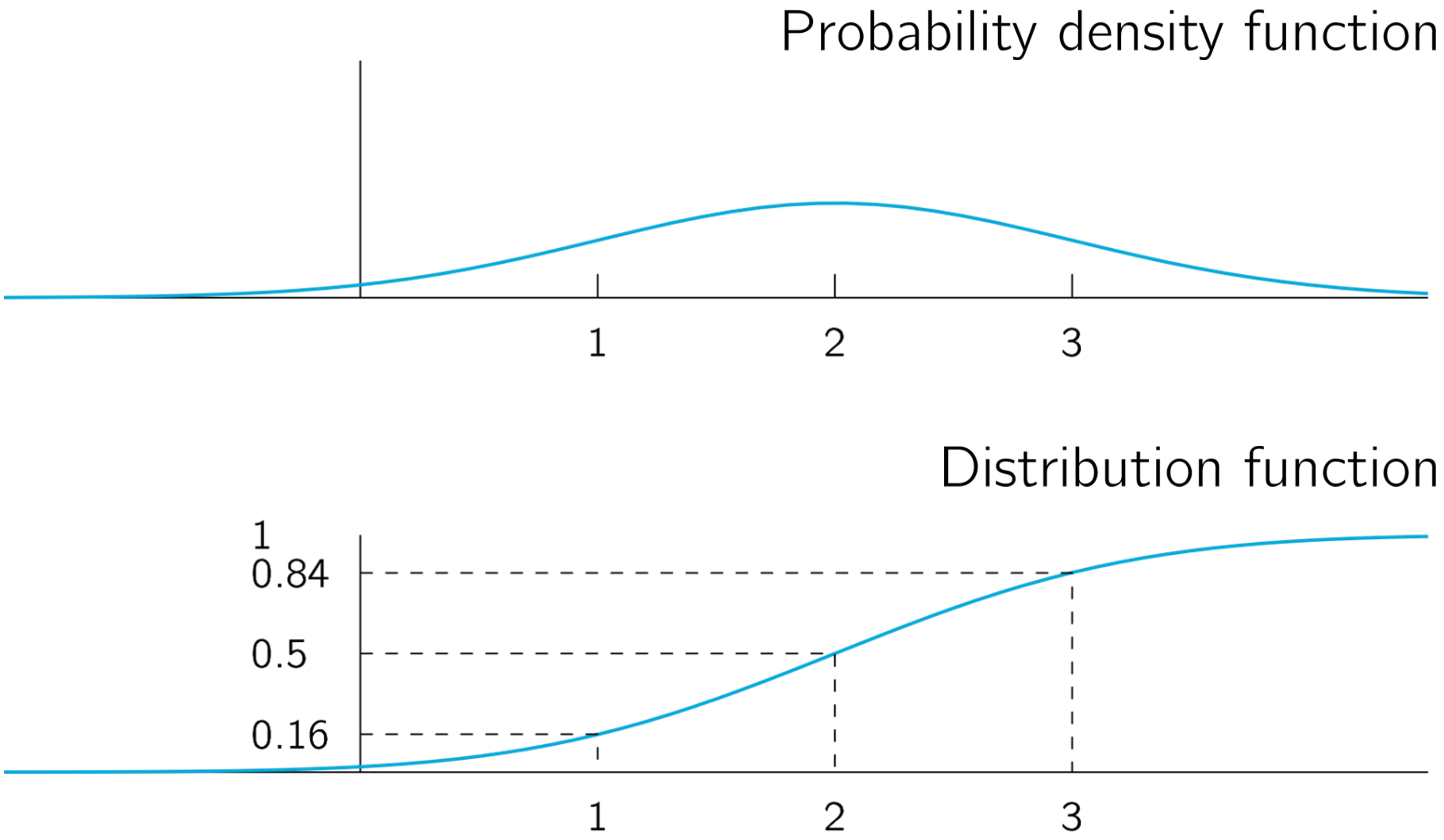
Definition:
A continuous random variable X has a normal distribution with parameters μ and σ^2 if its probability density function f is given by:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

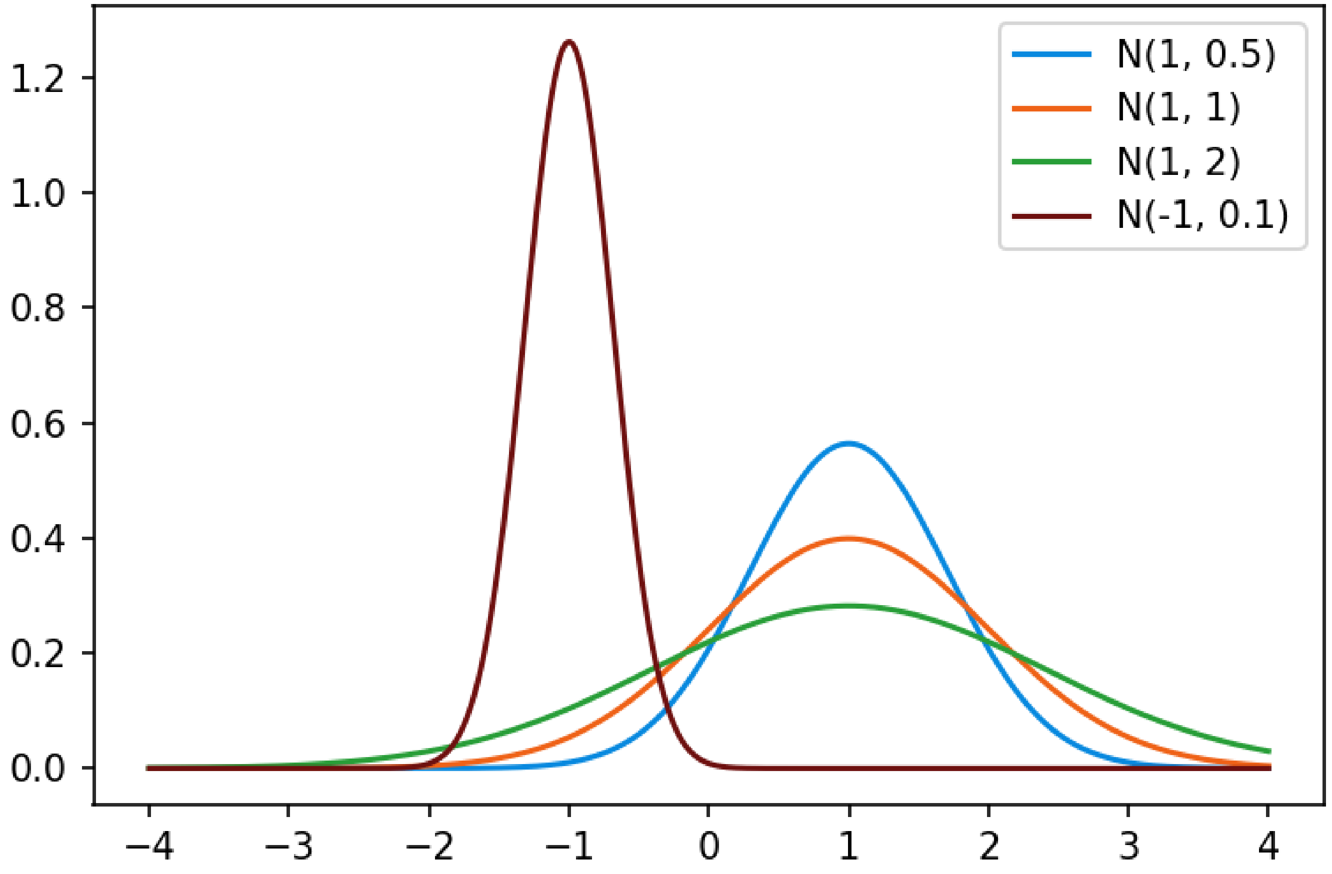
Notation: $X \sim N(\mu, \sigma^2)$.

Property: Symmetric around μ.

Normal distribution: $N(2,1)$



Normal distributions



The standard normal distribution

Definition:
If $\mu = 0$ and $\sigma^2 = 1$, the distribution $N(0,1)$ is called the standard normal distribution.

The right tail probabilities $P(Z \geq a)$ are given by the following normal table, which can be found on the Blackboard

Standard Normal Distribution Table (Right-Tail Probabilities)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641
0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121

Transform to standard normal distribution

- For a normal random variable X, the standard normal random variable can be transformed as

$$Z = (X - \mu) / \sigma$$

- The cumulative distribution function of X can be found by using:

$$\begin{aligned} F_Z(z) &= P(Z \le z) \\ &= P(\frac{X-\mu}{\sigma} \le z) \\ &= P(X \le \mu + z\sigma) \\ &= F_X(\mu + z\sigma) \end{aligned}$$

alternatively:

$$F_X(x) = F_Z(\frac{x-\mu}{\sigma})$$

Normal Distribution Example 1

An analog signal received at a detector (measured in microvolts) may be modeled as a Gaussian random variable N(200, 256) at a fixed point in time. What is the probability that the signal will exceed 240 microvolts? What is the probability that the signal is larger than 240 microvolts, given that it is larger than 210 microvolts?

Normal Distribution Example 2

Assuming that the life of a given subsystem, in the wear-out phase, is normally distributed with $\mu = 10{,}000$ hours and $\sigma = 1{,}000$ hours, determine the reliability for an operating time of 500 hours given that

- (a) The age of the component is 9,000 hours,
- (b) The age of the component is 11,000.

See you next week